

```
PS D:\folder\rac\ih\môn\hk1-4\ptht-phantan\LT\bt\b4\1> java Main
```

```
=====
CHUONG TRINH KIEM TRA HE THONG MESSAGE BROKER (BAI TAP 4)
=====
```

```
>>> 1. KIEM TRA 1.1: CONSUMER DANG CHAY <<<
```

```
[Customer] Sending ORDER-001
```

```
[Broker] Message received ORDER-001
```

```
[OrderService] Received ORDER-001
```

```
[OrderService] Completed ORDER-001
```

```
[Customer] Sending ORDER-002
```

```
[Broker] Message received ORDER-002
```

```
[OrderService] Received ORDER-002
```

```
[OrderService] Completed ORDER-002
```

```
[Customer] Sending ORDER-003
```

```
[Broker] Message received ORDER-003
```

```
[OrderService] Received ORDER-003
```

```
[OrderService] Completed ORDER-003
```

```
-----
>>> 2. KIEM TRA 1.2: CONSUMER CHUA CHAY (MESSAGE QUEUE) <<<
```

```
>>> 2. KIEM TRA 1.2: CONSUMER CHUA CHAY (MESSAGE QUEUE) <<<
```

```
--> Cho Customer gui message truoc (Chua chay OrderService):
```

```
[Customer] Sending ORDER-001
```

```
[Broker] Message received ORDER-001
```

```
[Customer] Sending ORDER-002
```

```
[Broker] Message received ORDER-002
```

```
[Customer] Sending ORDER-003
```

```
[Broker] Message received ORDER-003
```

```
--> Sau do moi khoi dong OrderService:
```

```
Ket qua:
```

```
[OrderService] Received ORDER-001
```

```
[OrderService] Received ORDER-002
```

```
[OrderService] Received ORDER-003
```

```
OrderService.java U Main.java U Customer.java U X
h > môn > hk1-4 > ptht-phantan > LT > bt > b4 > 1 > Customer.java
1 public class Customer {
2     private MessageBroker broker;
3
4     public Customer(MessageBroker broker) {
5         this.broker = broker;
6     }
7
8     /**
9     */
10    public void sendOrder(String orderId) {
11        System.out.println("[Customer] Sending " + orderId);
12        broker.sendMessage(orderId);
13    }
14 }
15
```

```
Customer.java U MessageBroker.java U InventoryService.java U x NotificationService.java U
iuh > môn > hk1-4 > ptht-phantan > LT > bt > b4 > 1 > InventoryService.java
1 public class InventoryService implements OrderSubscriber {
2     @Override
3     public void onOrderReceived(Order order) {
4         System.out.println("[InventoryService]");
5         System.out.println("Received " + order.getId());
6         System.out.println("Checking and updating inventory for product: " + order.getProduct());
7         try {
8             Thread.sleep(700); // Mô phỏng thời gian kiểm tra và cập nhật kho
9         } catch (InterruptedException e) {
10             Thread.currentThread().interrupt();
11         }
12         System.out.println("Inventory updated.");
13         System.out.println();
14     }
15 }
16
```

Main.java U × Customer.java U MessageBroker.java U InventoryService.java U Notific

ih > môn > hk1-4 > ptht-phantan > LT > bt > b4 > 1 > Main.java

```
1 public class Main {
2     public static void main(String[] args) {
3         System.out.println("=====");
4         System.out.println("    CHUONG TRINH KIEM TRA HE THONG MESSAGE BROKER (BAI TAP 4)    ");
5         System.out.println("=====\\n");
6
7         // -----
8         // KIEM TRA 1.1: CONSUMER DANG CHAY
9         // -----
10        System.out.println(">>> 1. KIEM TRA 1.1: CONSUMER DANG CHAY <<<");
11        MessageBroker broker1 = new MessageBroker();
12        OrderService orderService1 = new OrderService();
13        broker1.registerConsumer(orderService1);
14
15        Customer customer1 = new Customer(broker1);
16        customer1.sendOrder("ORDER-001");
17        customer1.sendOrder("ORDER-002");
18        customer1.sendOrder("ORDER-003");
19
20        System.out.println("\\n-----\\n");
21
22        // -----
23        // KIEM TRA 1.2: CONSUMER CHUA CHAY (TIME-UNCOUPLED)
24        // -----
25        System.out.println(">>> 2. KIEM TRA 1.2: CONSUMER CHUA CHAY (MESSAGE QUEUE) <<<");
26        MessageBroker broker2 = new MessageBroker();
27        Customer customer2 = new Customer(broker2);
28
29        System.out.println("--> Cho Customer gui message truoc (Chua chay OrderService):");
30        customer2.sendOrder("ORDER-001");
31        customer2.sendOrder("ORDER-002");
32        customer2.sendOrder("ORDER-003");
33
34        System.out.println("\\n--> Sau do moi khoi dong OrderService:");
35        System.out.println("Ket qua:");
36        OrderService orderService2 = new OrderService();
37        broker2.registerConsumer(orderService2);
38
39        System.out.println("\\n=====");
40    }
41 }
42
```

```
er.java U  MessageBroker.java U  InventoryService.java U  NotificationService.java U  Paym...  ← →
iuh > môn > hk1-4 > ptht-phantan > LT > bt > b4 > 1 > MessageBroker.java
1  import java.util.LinkedList;
2  import java.util.Queue;
3
4  public class MessageBroker {
5      private final Queue<String> messageQueue = new LinkedList<>();
6      private OrderService consumer = null;
7
8      public void sendMessage(String orderId) {
9          System.out.println("[Broker] Message received " + orderId);
10
11          if (consumer != null) {
12              consumer.processOrder(orderId);
13          } else {
14              messageQueue.add(orderId);
15          }
16      }
17
18      /**
19       */
20      public void registerConsumer(OrderService consumer) {
21          this.consumer = consumer;
22
23          // Xu ly cac message dang cho san trong hang doi
24          while (!messageQueue.isEmpty()) {
25              String pendingOrder = messageQueue.poll();
26              this.consumer.receivePendingOrder(pendingOrder);
27          }
28      }
29
30  }
```

```
er.java U  MessageBroker.java U  InventoryService.java U  NotificationService.java U  Paym...  ← → ...
iuh > môn > hk1-4 > ptht-phantan > LT > bt > b4 > 1 > NotificationService.java
1  public class NotificationService implements OrderSubscriber {
2      @Override
3      public void onOrderReceived(Order order) {
4          System.out.println("[NotificationService]");
5          System.out.println("Received " + order.getOrderId());
6          System.out.println("Sending notification...");
7          try {
8              Thread.sleep(600); // Mo phong thoi gian gui thong bao
9          } catch (InterruptedException e) {
10              Thread.currentThread().interrupt();
11          }
12          System.out.println("Notification sent.");
13          System.out.println();
14      }
15  }
16
```

NotificationService.java U Order.java U X PaymentService.java U OrderSubscriber.java U

uh > môn > hk1-4 > ptht-phantan > LT > bt > b4 > 1 > Order.java

```
1 public class Order {
2     private String orderId;
3     private String productName;
4     private double price;
5
6     public Order(String orderId, String productName, double price) {
7         this.orderId = orderId;
8         this.productName = productName;
9         this.price = price;
10    }
11
12    public String getOrderId() {
13        return orderId;
14    }
15
16    public String getProductName() {
17        return productName;
18    }
19
20    public double getPrice() {
21        return price;
22    }
23
24    @Override
25    public String toString() {
26        return orderId + " | " + productName + " | " + String.format("%.0f", price);
27    }
28 }
29
```

OrderService.java U X Main.java U Customer.java U MessageBroker.java U Inve

iuu > môn > hk1-4 > ptht-phantan > LT > bt > b4 > 1 > OrderService.java

```
1 public class OrderService {
2     /**
3      * Xu ly don hang khi Consumer dang chay (Kiem tra 1.1)
4      */
5     public void processOrder(String orderId) {
6         System.out.println("[OrderService] Received " + orderId);
7         System.out.println("[OrderService] Completed " + orderId);
8         System.out.println();
9     }
10
11     /**
12      * Nhan cac don hang ton dong trong Queue khi vua khoi dong (Kiem tra 1.2)
13      */
14     public void receivePendingOrder(String orderId) {
15         System.out.println("[OrderService] Received " + orderId);
16     }
17 }
18
```

NotificationService.java U Order.java U PaymentService.java U OrderSubscriber.java U ×

iuh > môn > hk1-4 > ptht-phantan > LT > bt > b4 > 1 > OrderSubscriber.java

```
1 public interface OrderSubscriber {
2     /**
3      * Nhan va xu ly don hang duoc phat tu Message Broker
4      * @param order Thông tin don hang
5      */
6     void onOrderReceived(Order order);
7 }
8
```

NotificationService.java U Order.java U PaymentService.java U × OrderSubscriber.java U

iuh > môn > hk1-4 > ptht-phantan > LT > bt > b4 > 1 > PaymentService.java

```
1 public class PaymentService implements OrderSubscriber {
2     @Override
3     public void onOrderReceived(Order order) {
4         System.out.println("[PaymentService]");
5         System.out.println("Received " + order.getOrderid());
6         System.out.println("Processing payment...");
7         try {
8             Thread.sleep(800); // Mo phong thoi gian xu ly thanh toan
9         } catch (InterruptedException e) {
10             Thread.currentThread().interrupt();
11         }
12         System.out.println("Payment completed.");
13         System.out.println();
14     }
15 }
16
```

```
Test1_1.java U X OrderService.java U Main.java U Customer.java U MessageBroker.java U
iuh > môn > hk1-4 > ptht-phantan > LT > bt > b4 > 1 > Test1_1.java
1 public class Test1_1 {
2     public static void main(String[] args) {
3         System.out.println("=== KIEM TRA 1.1: CONSUMER DANG CHAY ===");
4
5         MessageBroker broker = new MessageBroker();
6
7         OrderService orderService = new OrderService();
8         broker.registerConsumer(orderService);
9
10        Customer customer = new Customer(broker);
11        customer.sendOrder("ORDER-001");
12        customer.sendOrder("ORDER-002");
13        customer.sendOrder("ORDER-003");
14    }
15 }
16
```

```
Test2_2.java U X Test1_1.java U OrderService.java U Main.java U Customer.java U
iuh > môn > hk1-4 > ptht-phantan > LT > bt > b4 > 1 > Test1_2.java
1 public class Test1_2 {
2     public static void main(String[] args) {
3         System.out.println("=== KIEM TRA 1.2: CONSUMER CHUA CHAY ===");
4
5         MessageBroker broker = new MessageBroker();
6         Customer customer = new Customer(broker);
7
8         System.out.println("-> Customer gui 3 don hang vao Broker trong khi OrderService CHUA BAT:");
9         customer.sendOrder("ORDER-001");
10        customer.sendOrder("ORDER-002");
11        customer.sendOrder("ORDER-003");
12
13        System.out.println("\n--> Sau do moi khoi dong OrderService (Lay toan bo message ton trong Queue):");
14        OrderService orderService = new OrderService();
15        broker.registerConsumer(orderService);
16    }
17 }
18
```